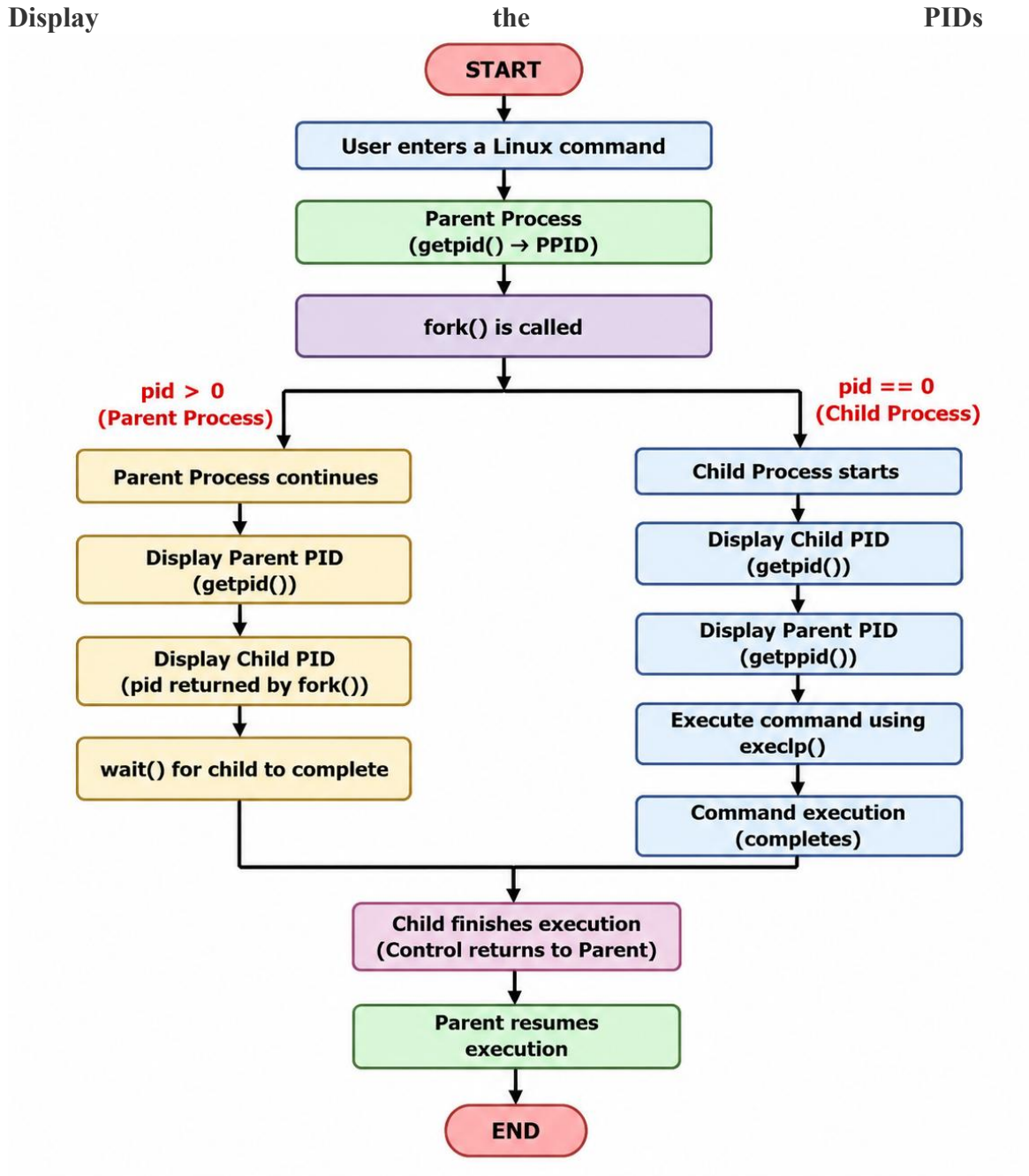


Task 1: Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

- a) **Write a C code to create a process using fork() system call.**
- b) **Read the user input as shell command.**
- c) **Use a system call exec() to execute the shell command.**
- d) **Use a wait() system call for waiting a parent process.**

e) Display

PIDs



of parent and child processes

Example Output:

Enter a Linux command: ls

Parent Process

Child Process

Parent PID : 67

Child PID : 68

Child PID : 68

Parent PID : 67

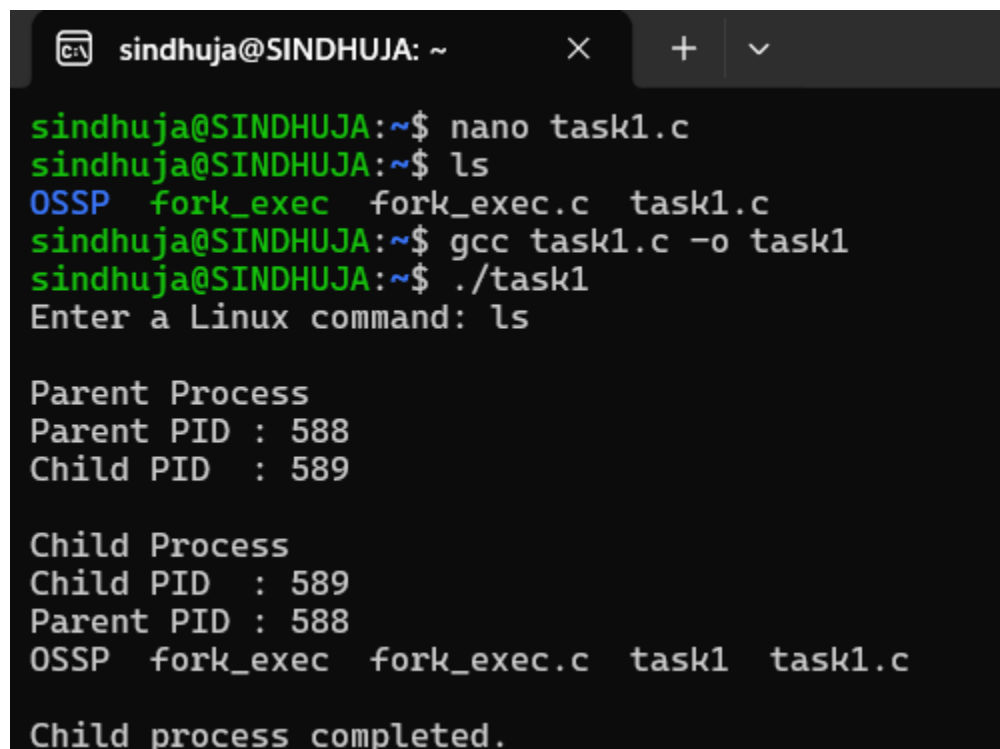
Executing command: ls

```
CE1  FIRST.c  File2.c    Proces_States.c  command_executor  filecopy.c
hello.c  process_example    session_4_a  session_4_b.c
```

```
CE1.c  File1.c  First.c    Process_States.c  command_executor.c  fork_example.c.save
my_shell  process_example.c  session_4_a.c
```

```
FIRST  File1.txt  Proces_States  R1          filecopy      hello
os-lab  process_example.cyes  session_4_b
```

Child process completed.



```
sindhuja@SINDHUJA: ~  
sindhuja@SINDHUJA:~$ nano task1.c  
sindhuja@SINDHUJA:~$ ls  
OSSP  fork_exec  fork_exec.c  task1.c  
sindhuja@SINDHUJA:~$ gcc task1.c -o task1  
sindhuja@SINDHUJA:~$ ./task1  
Enter a Linux command: ls  
  
Parent Process  
Parent PID : 588  
Child PID  : 589  
  
Child Process  
Child PID  : 589  
Parent PID : 588  
OSSP  fork_exec  fork_exec.c  task1  task1.c  
  
Child process completed.
```



sindhuja@SINDHUJA: ~



GNU nano 7.2

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main() {
    char command[100];

    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }
    else if (pid == 0) {
        // Child Process
        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        execlp(command, command, NULL);

        // Executes only if exec fails
        printf("Command execution failed!\n");
        exit(1);
    }
    else {
        // Parent Process
        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

```
^G Help
^X Exit
```

```
^O Write Out
^R Read File
```

```
^W Where Is
^_ Replace
```

```
^K Cut
^U Paste
```

```
^T Execute
^J Justify
```

Task-2: Using Linux terminal commands (uname, lscpu, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

uname -a

```
sindhuja@SINDHUJA: ~  
sindhuja@SINDHUJA:~$ uname -a  
Linux SINDHUJA 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYN  
AMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux  
sindhuja@SINDHUJA:~$ |
```

lscpu

```

sindhujas@SINDHUJA: ~$ uname -a
Linux SINDHUJA 6.18.35.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 x86_64 x86_64 GNU/Linux
sindhujas@SINDHUJA: ~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       12
On-line CPU(s) list:         0-11
Vendor ID:                    GenuineIntel
Model name:                   13th Gen Intel(R) Core(TM) i5-1335U
CPU family:                   6
Model:                        186
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                    1
Stepping:                     3
BogoMIPS:                     4991.99
Flags:                         fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtsc
                                cp lm constant_tsc rep_good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmulqdq vmx ssse3 fma cx16 pcid
                                sse4_1 sse4_2 x2apic movbe movbe_popt tsc_deadline_timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd ibrs i
                                bpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjust bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflush
                                opt clwb sha_ni xsaveopt xsavec xgetbv1 xsave_vnni vnni umip waitpkg gfni vaes vpclmulqdq rdpid movdiri movdir64b fsrm md_
                                clear serialize ibt flush_lld arch_capabilities

Virtualization features:
Virtualization:               VT-x
Hypervisor vendor:            Microsoft
Virtualization type:          full
Caches (sum of all):
L1d:                          288 KiB (6 instances)
L1i:                          192 KiB (6 instances)
L2:                            7.5 MiB (6 instances)
L3:                            12 MiB (1 instance)

```

```

NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Vulnerable: No microcode
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1: Mitigation; usercopy/swaps barriers and __user pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBR SB-eIBRS SW sequence; BHI BHI_DIS_S
  Srbds: Not affected
  Tsa: Not affected
  Tsx async abort: Not affected
  Vmscape: Not affected
sindhuja@SINDHUJA:~$ |

```

ps

```
sindhuja@SINDHUJA:~$ ps
  PID TTY          TIME CMD
  753 pts/2        00:00:00 bash
  779 pts/2        00:00:00 ps
sindhuja@SINDHUJA:~$ |
```

Top

```
sindhuja@SINDHUJA:~$ top
top - 05:15:14 up 39 min, 1 user, load average: 0.06, 0.03, 0.00
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7782.6 total, 7207.8 free, 526.2 used, 198.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7256.4 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21644	13044	9728	S	0.0	0.2	0:00.97	systemd
2	root	20	0	3180	2204	2072	S	0.0	0.0	0:00.02	init-systemd(Ub
6	root	20	0	3196	2132	2008	S	0.0	0.0	0:00.00	init
55	root	19	-1	50444	16308	15176	S	0.0	0.2	0:00.31	systemd-journal
103	root	20	0	25536	6864	5144	S	0.0	0.1	0:00.53	systemd-udev
114	systemd+	20	0	21460	13248	10996	S	0.0	0.2	0:00.12	systemd-resolve
115	systemd+	20	0	91028	7980	7012	S	0.0	0.1	0:00.07	systemd-timesyn
179	root	20	0	4244	2648	2396	S	0.0	0.0	0:00.01	cron
180	message+	20	0	9632	5344	4652	S	0.0	0.1	0:00.09	dbus-daemon
196	root	20	0	17972	8876	7840	S	0.0	0.1	0:00.10	systemd-logind
205	root	20	0	1756104	12020	9640	S	0.0	0.2	0:00.19	wsL-pro-service
237	root	20	0	3124	1976	1836	S	0.0	0.0	0:00.00	agetty
242	syslog	20	0	222516	6076	4676	S	0.0	0.1	0:00.12	rsyslogd
249	root	20	0	107020	23140	13672	S	0.0	0.3	0:00.16	unattended-upgr
367	root	20	0	6696	4656	3876	S	0.0	0.1	0:00.00	login
413	sindhuja	20	0	20312	11636	9516	S	0.0	0.1	0:00.12	systemd
414	sindhuja	20	0	21160	3580	1852	S	0.0	0.0	0:00.00	(sd-pam)
440	sindhuja	20	0	6080	5344	3676	S	0.0	0.1	0:00.02	bash
548	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
549	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.07	Relay(553)
553	sindhuja	20	0	6076	5416	3692	S	0.0	0.1	0:00.04	bash
748	root	20	0	3184	1116	980	S	0.0	0.0	0:00.00	SessionLeader
749	root	20	0	3200	1256	1108	S	0.0	0.0	0:00.01	Relay(753)
753	sindhuja	20	0	6076	5396	3672	S	0.0	0.1	0:00.02	bash
782	sindhuja	20	0	9308	5708	3512	R	0.0	0.1	0:00.01	top